

Ali HAJIABADI

Postdoctoral Researcher and Lecturer
ETH Zürich

CONTACT INFORMATION

ADDRESS: ETZ G 61.2, Gloriastr. 35, 8092 Zürich, Switzerland
EMAIL: ahajiabadi@ethz.ch
HOMEPAGE: ahajiabadi.github.io

RESEARCH INTERESTS

Systems Security, Formal Methods, Computer Architecture, Hardware/Software Co-design, Compilers, Secure Architectures and Software, Microarchitectural Side Channels, Machine Learning Systems Security

EDUCATION

- 2019 - 2024 Doctor of Philosophy in Computer Science
National University of Singapore (NUS), Singapore
Thesis: *"Building Efficient and Secure Processors thorough Hardware/Software Co-design"*
Advisor: Prof. Trevor E. CARLSON
- 2014 - 2019 Bachelor of Science in Computer Engineering
Sharif University of Technology, Tehran, Iran
Thesis: *"High Concurrency Latency Tolerant Register Files for GPUs"*
Advisor: Prof. Hamid SARBAZI-AZAD
- 2009 - 2013 Diploma in Physics and Mathematics Discipline
Shahid Beheshti High School, Birjand, Iran
Affiliated with the National Organization for the Development of Exceptional Talents (NODET)

HONORS & AWARDS

- JUL. 2026 NOTEWORTHY REVIEWER, as a PC member of EuroS&P '26.
- AUG. 2024 DEAN'S GRADUATE RESEARCH EXCELLENCE AWARD.
- JUL. 2024 SCHOOL OF COMPUTING CS PHD THESIS AWARD (Honorable Mention).
- JUN. 2024 BEST PAPER AWARD nomination at DAC '24.
- OCT. 2023 NUSGS RESEARCH INCENTIVE AWARD from School of Computing, NUS (S\$ 2,500 award).
- JAN. 2022 STUDENT TRAVEL AWARD from ASPLOS'22 conference.
- AUG. 2021 RESEARCH ACHIEVEMENT AWARD from School of Computing, NUS.
- MAR. 2020 Invited talk and travel grant for the 2nd Young Architect Workshop at ASPLOS'20, Switzerland.
- FEB. 2019 PRESIDENT'S GRADUATE FELLOWSHIP, the most prestigious doctoral fellowship in Singapore.
- SEP. 2014 Ranked 164th in Iranian National University Entrance Exam among more than 250,000 students.
- 2006/2009 Recognized as talented student in entry exam of NODET for middle school and high school.

PROFESSIONAL EXPERIENCE

- JUL. 2024 - PRESENT Postdoctoral Researcher at ETH ZÜRICH, Switzerland
Computer Security (COMSEC) Group
Group Lead: Prof. Kaveh RAZAVI
Research: CPU and DRAM security, novel attacks and defenses for computing systems
- AUG. 2019 - JUN. 2024 Graduate Research Assistant at NATIONAL UNIVERSITY OF SINGAPORE, Singapore
NUS Computer Architecture (CompArch) Group
Group Lead: Prof. Trevor E. CARLSON
Research: hardware/software co-design for efficient and secure modern processors
- JUL. 2016 - JUN. 2019 Research Assistant at SHARIF UNIVERSITY OF TECHNOLOGY, Tehran, Iran
High Performance Computer Architectures and Networks (HPCAN) Lab
Group Lead: Prof. Hamid SARBAZI-AZAD
Research: GPU register prefetching via HW/SW co-design and compiler optimizations
- SUMMER 2018 Research Intern at NATIONAL UNIVERSITY OF SINGAPORE, Singapore
Group Lead: Prof. Trevor E. CARLSON
Research: exploring new implementations for out-of-order commit processors

PEER-REVIEWED PUBLICATIONS

17. **Ali Hajiabadi**, Silvan Niederer, Kaveh Razavi
CCS'26 *VeriPRAC: Verifiable Formal Contracts for Secure Per Row Activation Counting.*
Proceedings of 33rd ACM Conference on Computer and Communications Security (CCS 2026), November 2026.
16. Silvan Niederer, **Ali Hajiabadi**, Kaveh Razavi
DIMVA'26 *Glasswall: Enforcing User-Defined Data Usage Policies in Functions as a Service.*
Proceedings of 23rd Conference on Detection of Intrusions and Malware & Vulnerability Assessment, July 2026. Acceptance rate: 33/151 = 21.8%.
15. Jean-Claude Graf*, Sandro Rüegg*, **Ali Hajiabadi**, Kaveh Razavi
S&P'26 *VMscape: Exposing and Exploiting Incomplete Branch Predictor Isolation in Cloud Environments.*
Proceedings of 47th IEEE Symposium on Security and Privacy (S&P, Oakland 2026), May 2026. Acceptance rate: 118/925 = 12.8%.
14. Hritvik Taneja, **Ali Hajiabadi**, Michele Marazzi, Kaveh Razavi, Moinuddin K. Qureshi
HPCA'26 *MIRZA: Efficiently Mitigating Rowhammer with Randomization and ALERT.*
Proceedings of 32nd IEEE International Symposium on High-Performance Computer Architecture (HPCA 2026), February 2026. Acceptance rate: 119/602 = 19.8%.
13. **Ali Hajiabadi**, Michele Marazzi, Kaveh Razavi
CCS'25 *CHaRM: Checkpointed and Hashed Counters for Flexible and Efficient Rowhammer Mitigation.*
Proceedings of 32nd ACM Conference on Computer and Communications Security (CCS 2025), October 2025. Acceptance rate: 316/2186 = 14.5%.
12. Silvan Niederer, Sandro Rüegg, **Ali Hajiabadi**, Kaveh Razavi
MICRO'25 *One Flew over the Stack Engine's Nest: Practical Microarchitectural Attacks on the Stack Engine.*
Proceedings of 58th IEEE/ACM International Conference on Microarchitecture (MICRO 2025), October 2025. Acceptance rate: 124/597 = 20.8%.
11. **Ali Hajiabadi**, Trevor E. Carlson
ISCA'25 *CASSANDRA: Efficient Enforcement of Sequential Execution for Cryptographic Programs.*
Proceedings of 52nd ACM/IEEE International Symposium on Computer Architecture (ISCA 2025), June 2025. Acceptance rate: 132/570 = 23.1%.
10. Yun Chen*, **Ali Hajiabadi***, Romain Poussier, Yaswanth Tavva, Andreas Diavastos, Shivam Bhasin, Trevor E. Carlson
TACO'25 *PARADISE: Criticality-Aware Instruction Reordering for Power Attack Resistance.*
In ACM Transactions on Architecture and Code Generation (TACO), 2024.
*Joint first-authors with equal contribution.
9. **Ali Hajiabadi**, Trevor E. Carlson
DAC'24 *CONJURING: Leaking Control Flow via Speculative Fetch Attacks.*
Proceedings of 61st ACM/IEEE Design Automation Conference (DAC 2024), June 2024. Acceptance rate: 337/1465 = 23.0%.
★ *Best Paper Award Nominee* (5/337 = 1.5%).
8. **Ali Hajiabadi**, Archit Agarwal, Andreas Diavastos, Trevor E. Carlson
DAC'24 *LEVISO: Efficient Compiler-Informed Secure Speculation.*
Proceedings of 61st ACM/IEEE Design Automation Conference (DAC 2024), June 2024. Acceptance rate: 337/1465 = 23.0%.
7. Yun Chen*, **Ali Hajiabadi***, Trevor E. Carlson
HPCA'24 *GADGETSPINNER: A New Transient Execution Primitive using the Loop Stream Detector.*
Proceedings of 30th IEEE International Symposium on High-Performance Computer Architecture (HPCA 2024), March 2024. Acceptance rate: 75/410 = 18.3%.
*Joint first-authors with equal contribution.
6. Yun Chen, **Ali Hajiabadi**, Lingfeng Pei, Trevor E. Carlson
HPCA'24 *PREFETCHX: Cross-Core Cache-Agnostic Prefetcher-Based Side-Channel Attacks.*
Proceedings of 30th IEEE International Symposium on High-Performance Computer Architecture (HPCA 2024), March 2024. Acceptance rate: 75/410 = 18.3%.
5. Arash Pashrashid, **Ali Hajiabadi**, Trevor E. Carlson
ICCAD'23 *HIDFIX: Efficient Mitigation of Cache-based Spectre Attacks through Hidden Rollbacks.*
Proceedings of 42nd IEEE/ACM International Conference on Computer-Aided Design (ICCAD 2023), November 2023. Acceptance rate: 172/768 = 22.4%.

4. ICCAD'22	Arash Pashrashid, Ali Hajiabadi , Trevor E. Carlson <i>Fast, Robust and Accurate Detection of Cache-based Spectre Attack Phases.</i> Proceedings of 41 st IEEE/ACM International Conference on Computer-Aided Design (ICCAD 2022), November 2022. Acceptance rate: 132/586 = 22.5%.
3. ASPLOS'21	Ali Hajiabadi , Andreas Diavastos, Trevor E. Carlson <i>NOREBA: A Compiler-Informed Non-speculative Out-of-Order Commit Processor.</i> Proceedings of 26 th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2021), April 2021. Acceptance rate: 75/398 = 18.8%.
2. TOCS'21	Mohammad Sadrosadati, Amirhossein Mirhosseini, Ali Hajiabadi , Seyed Borna Ehsani, Hajar Falahati, Hamid Sarbazi-Azad, Mario Drumond, Babak Falsafi, Rachata Ausavarungnirun, Onur Mutlu <i>Highly Concurrent Latency-tolerant Register Files for GPUs.</i> In ACM Transactions on Computer Systems (TOCS), 2021.
1. CGO'21	Harish Patil, Alexander Isaev, Wim Heirman, Alen Sabu, Ali Hajiabadi , Trevor E. Carlson <i>ELFies: Executable Region Checkpoints for Performance Analysis and Simulation.</i> Proceedings of 19 th IEEE International Symposium on Code Generation and Optimization (CGO 2021), March 2021. Acceptance rate: 31/89 = 34.8%.

TEACHING EXPERIENCE

► ETH Zürich, Switzerland

FALL 2026	Lecturer , Computer Security (BSc) [New: RISC-V edition]
SPRING 2026	Co-lecturer , Computer Engineering (BSc), with Prof. Kaveh Razavi and Dr. Patrick Jattke
FALL 2025	Lecturer , Computer Security (BSc) [New Course]
SPRING 2025	Lecturer , Capture the Flag – Introduction to Cybersecurity (P&S BSc) [New Course]
SPRING 2025	Co-lecturer , Computer Engineering (BSc), with Prof. Kaveh Razavi

► National University of Singapore, Singapore

SPRING 2020	Teaching Assistant (tutorial instructor), CS2106 Introduction to Operating Systems (BSc)
& SPRING 2021	Lecturer : Prof. Djordje Jevdjic

► Sharif University of Technology, Tehran, Iran

SPRING 2017	Teaching Assistant (assignments & projects), CE323 Computer Architecture (BSc) <i>Lecturer: Prof. Hamid Sarbazi-Azad</i>
FALL 2017	Teaching Assistant (tutorial instructor, assignments & projects), CE453 Real-Time Systems (BSc)
& FALL 2018	<i>Lecturer: Prof. Amirhossein Jahangir</i>

PROFESSIONAL SERVICES

Program Committee	S&P ('27), uASC ('27, '26), HPCA ('27, '26), MICRO ('26, '25), ISCA ('26), EuroS&P ('26), AsiaCCS ('26)
Sub-reviewer	USENIX Security ('25)
Reviewer (journals)	IEEE CAL, ACM TACO, ACM TRETs
Session Chair	uASC'26 (<i>Brave New Defense</i>), MICRO'25 (<i>Security and Privacy - Memory</i>)
Shadow PC	EuroSys ('24, '23)
Student Volunteer	PLDI ('21)

TALKS

NOV. 2025	The Hidden and Ugly Faces of Spectre: Unexplored and Unmitigated Threat Models <i>IC, EPFL, Lausanne, Switzerland.</i>
OCT. 2025	CHaRM: Checkpointed and Hashed Counters for Flexible and Efficient Rowhammer Mitigation <i>Conference on Computer and Communication Security (CCS 2025), Taipei, Taiwan.</i>
JUN. 2025	CASSANDRA: Efficient Enforcement of Sequential Execution for Cryptographic Programs <i>Int. Symposium on Computer Architecture (ISCA 2025), Tokyo, Japan.</i>
MAR. 2024	Will CPUs Be Free of Spectre? Dark Side and Light Side of the Battle <i>ETH Zürich, COMSEC Group, Zürich, Switzerland.</i>
MAR. 2024	GADGETSPINNER: A New Transient Execution Primitive using the Loop Stream Detector <i>Int. Symposium on High-Performance Computer Architecture (HPCA 2024), Edinburgh, Scotland.</i>

AUG. 2021	NOREBA: A Compiler-Informed Non-speculative Out-of-Order Commit Processor <i>Computing Research Week, School of Computing (NUS), Virtual.</i>
APR. 2021	NOREBA: A Compiler-Informed Non-speculative Out-of-Order Commit Processor <i>Int. Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2021), Virtual.</i>
FEB. 2021	Accelerating HPC applications with Out-of-Order Commit Processors <i>Free and Open source Software Developers' European Meeting (FOSDEM 2021), HPC, Big Data, and Data Science track, Virtual.</i>
MAR. 2020	Speculation-Free Out-of-Order Commit <i>2nd Young Architect Workshop at the 25th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2020), Virtual.</i>

RESEARCH MENTORING

MAR. 2026 - PRESENT	Yves Rüetschi	BSc thesis, ETH Zürich	
FEB. 2026 - PRESENT	Tomas Jossen	BSc thesis, ETH Zürich	
FEB. 2026 - PRESENT	Jonas Buchholz	MSc thesis, ETH Zürich	Embargo
JAN. 2026 - PRESENT	Quentin Bordier	PhD thesis, COMSEC, ETH Zürich	Embargo
MAR. 2025 - MAY 2026	Jean-Claude Graf	PhD thesis, COMSEC, ETH Zürich	1 publication (SGP'26)
MAR. 2025 - MAY 2026	Sandro Rüegge	PhD thesis, COMSEC, ETH Zürich	1 publication (SGP'26)
MAR. 2025 - SEP. 2025	Maximilian Mosler	MSc thesis, ETH Zürich	Embargo
MAR. 2025 - JUN. 2025	Jonas Buchholz	Semester project, ETH Zürich	Embargo
DEC. 2024 - JUN. 2025	Chenfei Liu	MSc thesis, ETH Zürich	1 publication (ASPLOS'27)
SEP. 2024 - FEB. 2025	Mounir Raki	Semester project, ETH Zürich	
JUL. 2024 - PRESENT	Silvan Niderer	PhD thesis, COMSEC, ETH Zürich	2 publications (MICRO'25, DIMVA'26)
JAN. 2021 - DEC. 2023	Arash Pashrashid	PhD thesis, NUS CompArch	2 publications (ICCAD'23, ICCAD'22)
JUL. 2021 - JUL. 2023	Archit Agarwal	RA, NUS CompArch	1 publication (DAC'24)
SEP. 2020 - SEP. 2021	Vernon Pang	BSc final-year project, NUS	
AUG. 2020 - MAR. 2024	Yun Chen	PhD thesis, NUS CompArch	3 publications (2×HPCA'24, TACO)

SKILLS

PROGRAMMING LANGUAGES:	C, C++, Python, bash, and familiar with Java, Matlab, Scala
INSTRUCTION SET ARCHITECTURES:	x86, Arm, RISC-V
SCIENTIFIC TOOLS:	LLVM Compiler Infrastructure, gem5 Simulator, Sniper Simulator, Ramulator, Intel Pin, DynamoRIO